

Carbon-Aware GPU Jobs Scheduling via Local Search methods

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1 Definition of the problem

This work formulates job scheduling as a joint spatial-temporal optimization problem, where each job is assigned a start time and execution region to minimize a residual-load-based cost, used as a proxy for carbon intensity. Lower residual load corresponds to periods with higher renewable energy availability, making them preferable for carbon-efficient execution. Under this formulation, the scheduling cost accumulates over the execution window of each job and scales with both runtime and GPU demand, thereby penalizing large workloads executed during carbon-intensive periods [3]. The resulting combinatorial search problem is addressed using multiple local and population-based optimization algorithms under a fixed evaluation budget. This approach enables a systematic comparison of scheduling strategies and provides insights into the effectiveness of algorithmic decision-making in reducing the environmental impact of AI workloads across heterogeneous energy systems.

2 Datasets

For energy data we used the **Time Series** data package, curated by the Open Power System Data platform, which provides a harmonized collection of power system variables across 32 European countries for the period 2015 to mid-2020 [1]. This dataset aggregates hourly resolution data, with 15-minute and 30-minute resolutions available for specific regions, encompassing electricity consumption (load), wind and solar generation, installed capacities, and day-ahead electricity prices. For this study, we specifically utilized data from the **Blue Waters** cluster at NCSA, which consists of 26,864 nodes, 396,000 cores, and 4,228 GPUs [2]. The dataset includes 15,000 jobs and contains critical performance metrics, including unique job identifiers, CPU and GPU requirements, arrival intervals, and job completion status.

3 Methodology

3.1 Objective Function

The residual load, $RL(t)$, represents the electricity demand unmet by non-dispatchable renewable sources, where $D(t)$ is demand, $S(t)$ is solar, and $W_{\text{on}}(t)$ and $W_{\text{off}}(t)$ are onshore and offshore wind generation. To guide the optimization across different regions, this is converted into a dimensionless **Normalized Residual Load** (NRL), which scales the unmet demand relative to each country's actual grid capacity:

$$RL(t) = D(t) - [S(t) + W_{\text{on}}(t) + W_{\text{off}}(t)] \quad \text{NRL} = \begin{cases} \frac{\max(0, RL)}{\text{Load}} & \text{if Load} > 0 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

This formulation improves upon the preliminary cost function from earlier project phases, which caused unrealistic allocations by ignoring absolute regional capacities. By incorporating true grid constraints, the updated objective function prevents over-allocating jobs to low-capacity regions, resulting in a realistic and feasible energy reduction of approximately 19%. The scheduling objective minimizes cumulative residual load over a job's execution window:

$$\text{cost}(s, r, j) = \left(\sum_{t=s}^{s+\text{run_time}_j-1} \text{NRL}_r(t) \right) \cdot \frac{\text{gpu_num}_j}{3600} \quad (2)$$

where s is the start time in seconds, r is the assigned region, j is the job, and $RL_r(t)$ is the per-second residual load signal for region r . Giving the cost the interpretation of GPU-weighted residual energy exposure over the job’s execution window. Any candidate whose execution window extends beyond the available signal is assigned a cost of infinity and immediately discarded. All experiments operate over a contiguous 24-hour window. The default window begins at hour index 100 and spans 24 hours. Jobs whose run time exceeds 86,400 seconds (24 hours) are excluded from the benchmark.

3.2 Neighbourhood

Neighbor generation combines spatial and temporal perturbations: with a probability of $p = 0.35$, a region switch is performed; otherwise, a time shift is applied according to:

$$s \leftarrow \text{clamp}(s + \delta, 0, s_{\max}), \quad \delta \sim U(-20000, 20000) \quad (3)$$

The time move radius of 20,000 seconds was chosen to span roughly a quarter of a 24-hour scheduling window at second resolution, balancing exploration range against the risk of large aimless jumps. The spatial switch probability of 0.35 ensures that cross-region moves occur frequently enough to allow for meaningful exploration of spatial dimensions, while the majority of steps are allocated to temporal analyses within the currently identified best region.

3.3 Algorithms - AI tools used

Random, Hill Climbing, Local Beam Search, Simulated Annealing (with geometric cooling), Tabu Search, and Genetic Algorithm are evaluated under a budget of 80 evaluations for the sake of fairness. These algorithms are directly related to the syllabus of our course.

3.4 Spatial Ranking

For a given region r and a set of algorithms A , the weighted score $WS(r)$ and the algorithm reduction are defined on the same line as:

$$WS(r) = \frac{\sum_{a \in A} \text{share}(a, r) \cdot \max(\text{reduction}(a), 0)}{\sum_{a \in A} \max(\text{reduction}(a), 0)} \quad \text{reduction}(a) = \frac{C_{\text{random}} - C_{\text{alg}}}{C_{\text{random}}} \quad (4)$$

Algorithms yielding zero or negative reduction are excluded from the weight computation by being clamped to zero. Regions are then ranked in descending order of their weighted score, utilizing the mean share as a tiebreaker.

4 Results and Conclusion

As illustrated in Table 4, **Local Beam Search** demonstrated the highest efficacy, achieving a 19.37% reduction. The results indicate a highly competitive performance among the top four methods, with Simulated Annealing (19.27%), Tabu Search (19.22%), and Hill Climbing (19.19%) all falling within a narrow margin of 0.18% of the leading algorithm. Conversely, the Genetic Algorithm exhibited the lowest relative improvement at 17.76%, underperforming the other local search approaches by a notable margin. These results suggest that for the specific constraints of spatial residual-load scheduling, local-search-based metaheuristics offer a slight but statistically visible advantage over evolutionary approaches.

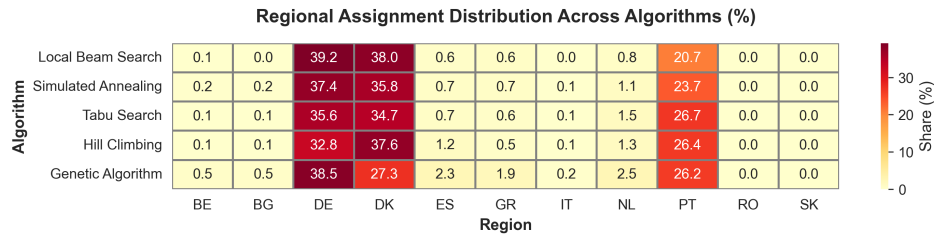


Figure 1: Countries preferred by the algorithms

Algorithm	Reduction vs. Random Baseline (%)
Local Beam Search	19.37%
Simulated Annealing	19.27%
Tabu Search	19.22%
Hill Climbing	19.19%
Genetic Algorithm	17.76%

Table 1: Algorithm Performance Summary

On the other hand, the preferred countries by the algorithms are shown in heatmap 4. Germany, Denmark and Portugal are selected most by the algorithms that shows a huge potential and usage of renewable energy are observed for data centers.

4.1 Good for Society

This project aligns with the **AI for Good** concept by using optimization to reduce the environmental impact of modern AI systems. As AI workloads grow, data centers consume massive amounts of electricity and drive up global carbon emissions [4]. This project addresses the issue by scheduling GPU jobs during periods of cleaner energy and in regions with higher renewable availability, lowering carbon emissions without needing to modify the AI models themselves. By combining optimization algorithms with real-world energy data, this work supports green data center operations and sustainable cloud infrastructure. Ultimately, it demonstrates how AI technologies can be leveraged not just for performance, but for meaningful environmental and societal benefit.

References

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